Appendix B

These are the translations of the original instructions, which were in German. The originals are available upon request from the authors.

Introduction

Welcome to the AWI Lab. In today's experiment, you will participate in 15 auctions. The experiment will take about one hour, and at the end of the experiment, you will be paid in cash. The payment you receive for the experiment depends on your own decisions, the decisions of the other participants, and on chance.

You can make all your choices at your computer. Please do not talk to other participants. If you have any questions, please raise your hand, and someone will come over.

Before we start the experiment, you are required to answer a few test questions correctly to make sure that you have understood the instructions. Now please read the instructions carefully. You may use paper and pencil and take notes any time.

Instructions

Groups

Before the experiment starts, all participants are divided into groups of four. These groups remain unchanged throughout the experiment. You will interact only with the members of your group, but not with the members of other groups.

What is the value of a good for an individual bidder?

The experiment consists of 15 auctions. In each group at each auction a good is sold to one of the 4 bidders. Prior to each one of the 15 auctions, a value for the good is determined randomly for each bidder. This value is an integer between 0 and 100. The value is randomly generated by the computer, and each integer is selected with the same probability.

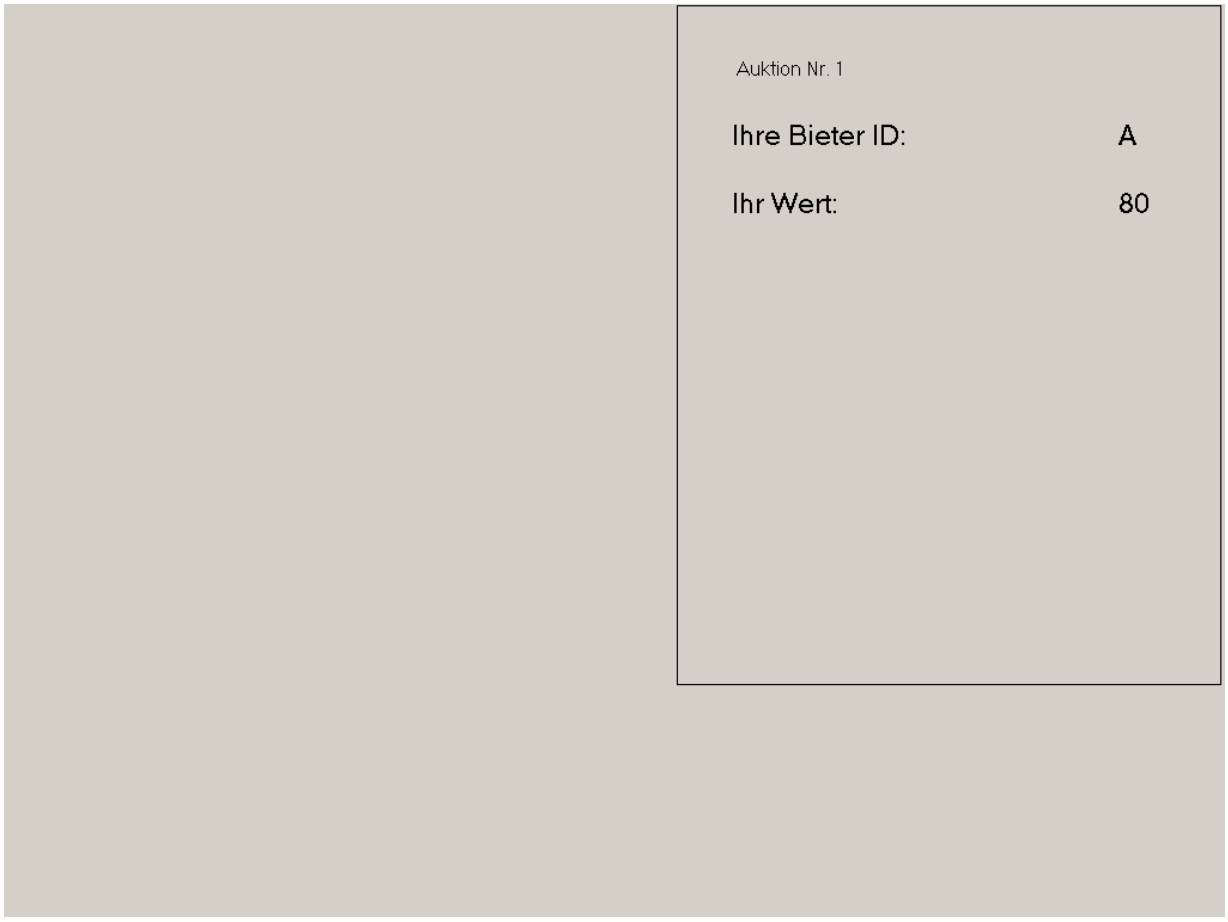
The value is determined independently for each bidder. The values of other bidders have no effect on the value determined for you. The value for each of the 15 auctions is also determined independently. The values in other auctions have no effect on the values determined in the current auction. Therefore your values for the good are most likely different from the values of other bidders in your group. Additionally, you most likely do not have the same value for all 15 auctions.

Each bidder knows his/her own value, but not the values of other bidders.

Screen 1: Value and Bidder ID

Prior to each of the 15 auctions, you will be shown your value for 5 seconds. You will see the screen shown in Figure 1. This screen also shows your Bidder ID. This ID serves as an identification of the different bidders within a group. Your Bidder ID is randomly determined for each of the 15 auctions. Therefore you and any other participant are unlikely to be assigned the same Bidder ID each time.

Figure 1: Screen 1:



{Translation: Auction No. 1

Your bidder ID: A

Your value: 80}

Auction procedure

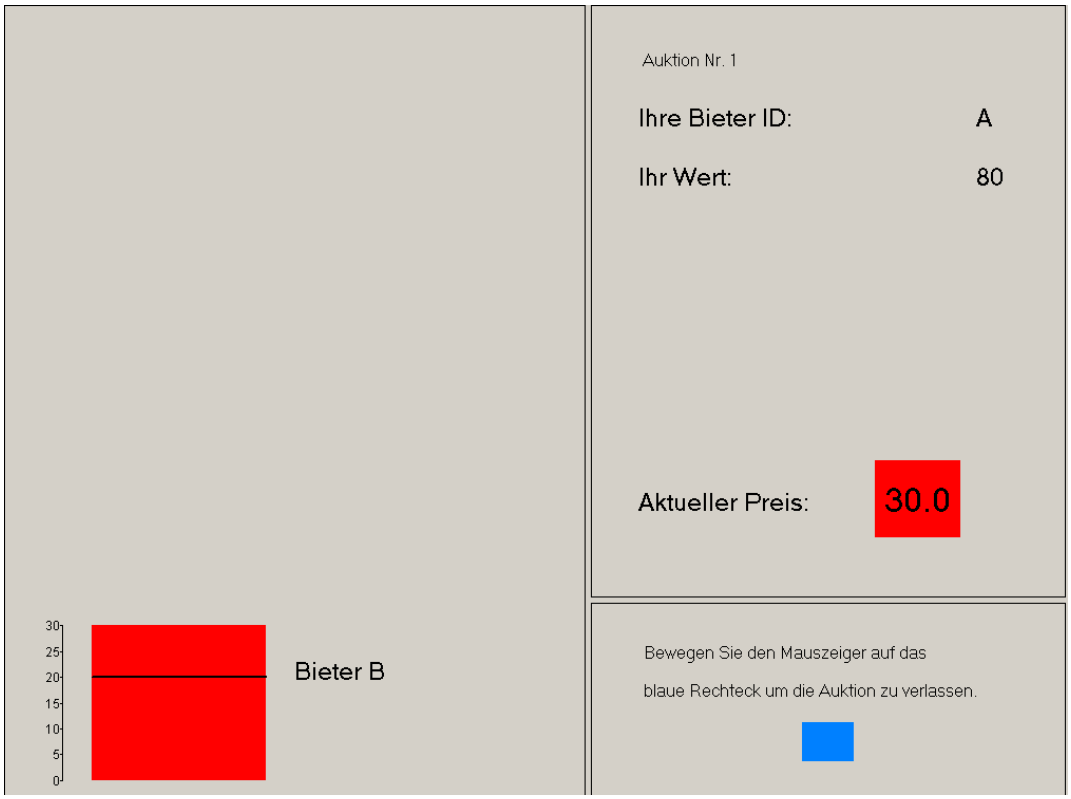
At the start of each auction, the price of a good is 0. Every 0.5 seconds, the price is raised by one point. Once the price has reached 34 points, the price will be raised only every second by one point.

You can decide to drop out of the auction any time by moving the cursor onto the blue rectangle at the right bottom on your screen (see Figure 2). Clicking the mouse is not required. Once you have moved the cursor onto the blue rectangle, you drop out of the auction.

Once 3 or 4 bidders of your group have dropped out, the auction is finished. The bidder who did not drop out purchases the good at the current price. If several bidders drop out simultaneously as last bidders in an auction, the buyer will be determined randomly: all of the bidders dropped out last have the same chance to buy the good at the current price.

Please look at Figure 2. On the left part of your screen, the red bar shows the current price. You can also see when another bidder of your group drops out of the auction. In Figure 2, for example, bidder B dropped out at price 20. The right part of the screen shows your Bidder ID, your value, and the current price (in a red field).

Figure 2: Screen 2



{Translation: Auction No. 1, Your Bidder ID: A, Your Value: 80, Current price: 30.0 Bidder B, "Please move the mouse pointer to the blue rectangle to leave the auction"}

End of Auction

When you move the cursor onto the blue rectangle, you will be notified that you have dropped out of the auction. Figure 3, for example, shows that bidder C dropped out at price 40. When 3 or more bidders of your group have dropped out, the auction is finished.

When the current price has reached 150 points, the auction is finished, even if fewer than 3 bidders have dropped out of the auction. The good will be sold at a price of 150 points to one of the remaining bidders, with the same probability for each bidder.

Figure 3: Screen 3



{Translation: Auction No. 1, Your Bidder ID: C, Your Value: 40, "You have left the auction." Current Price: 41.0. Bidder B, Bidder C, "Please move the mouse pointer to the blue rectangle to leave the auction"}

Calculating your profit

The bidder who buys the good receives a profit equal to the difference between the value of the good and the sales price. Additionally the buyer and the bidder who dropped out third (that is, the next to last remaining bidder) receive a bonus. This means you can make a profit in this auction even if you do not buy the good.

For calculating this bonus, two prices are relevant: the sales prices and the so-called basis price. The basis price is the price at which the second bidder drops out. Assume, for example, that bidder B drops out at price 20. Then bidder C drops out at price 40. So in this example the basis price is 40. The basis price is indicated by a thick blue line (see Figure 4).

The bonus equals half the difference between sales price and basis price, and is therefore always positive. Example: Bidder B drops out at price 20. Bidder C drops out at price 40. The basis price is therefore 40. Bidder D drops out at price 50. So the sales price is 50. In this case the bonus is 5: (sales price 50 – basis price 40)/2. Bidders B and C receive a profit of 0. Bidder D dropped out third and receives a profit of 5 (the bonus). Bidder A purchases the good. His/her profit equals the difference between the sales price of 50 and the value of the good plus the bonus of 5.

Figure 4: Screen 4



Your profit for each auction is calculated as follows:

- If you drop out first or second: 0
- If you drop out third: bonus = (sales price – basis price)/2
- If you purchase the good: your value – sales price + bonus

We consider another example: Assume that bidder A drops out at price 20. Bidders B and C drop out simultaneously at price 30. Since only one bidder stays, the auction ends, and bidder D buys the good at price 30. As the second bidder dropped out at price 30, the basis price is 30. So in this case the bonus is 0. Bidders A, B, and C gain a profit of 0, and bidder D gains a profit equaling the difference between his value of the good and the sales price 30.

We consider a final example to illustrate the calculation of profits. Assume that the values of bidders A – D are allocated in the following manner:

- Bidder A: 10
- Bidder B: 30
- Bidder C: 50
- Bidder D: 70

We assume that bidders A - D will drop out at the following prices:

- Bidder A: 15
- Bidder B: 50
- Bidder C: 80
- Bidder D: 80

The good is purchased at sales price 80 by bidder 3 or bidder 4, with equal probabilities. Since the second bidder dropped out at price 50, the basis price is 50. Therefore the bonus is 15 (sales price 80 – basis price 50)/2. The two last remaining bidders, that is, bidders C and D, receive a bonus of 15 each.

Now we look at the profit of bidder C. If bidder C buys the good, he/she pays the sales price of 80 for the good. He/she gets a value of 50 plus the bonus of 15, thus suffering a loss of 15 (value 50 – sales price 80 + bonus 15 = profit -15). As bidders C and D dropped out at price 80, each of them buys the good with a probability of 0.5. If bidder D buys the good, the profit of bidder C is 15: He/she only gets the bonus.

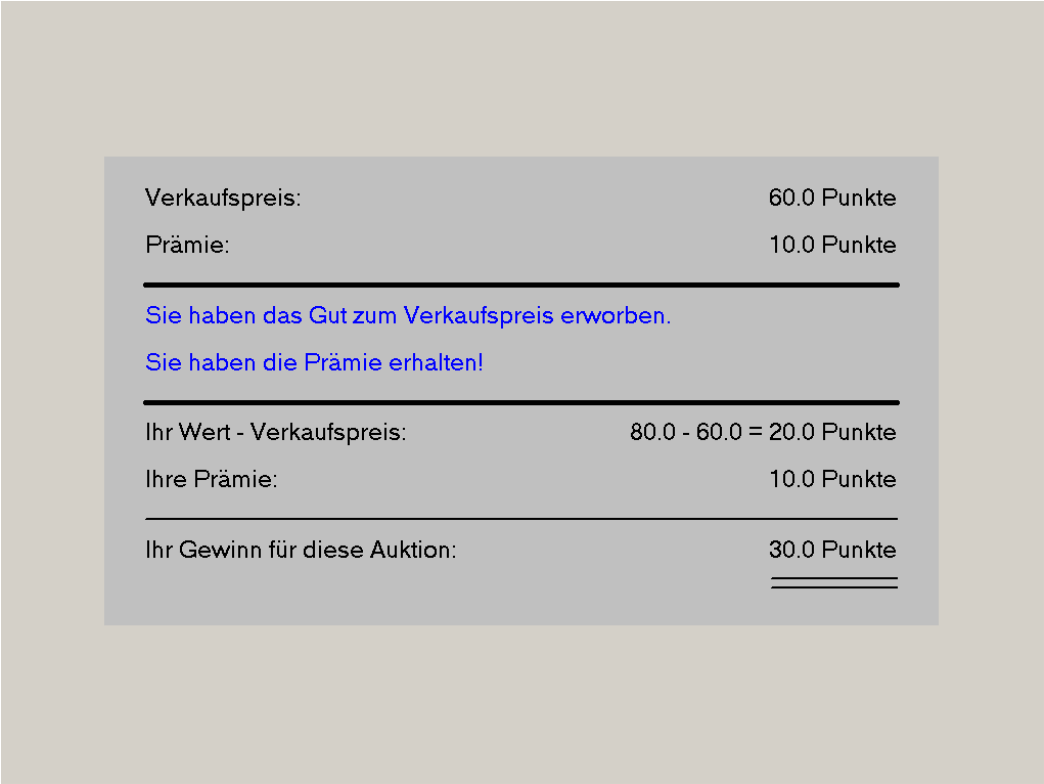
Display of Profits

At the end of the auction, the results of the auction are displayed for 15 seconds (see Figure 5). At the top you can see the sales price and the bonus. Next, in the middle of the screen, you learn whether you have purchased the good and whether you received the bonus. If you have purchased the good or received the bonus, this information is printed blue. It is printed *black* if you have *not* purchased the good or *not* received the bonus.

In the bottom part of the screen your profits are presented in detail. If you have purchased the good, the difference between your value and the sales price is displayed. If you did not purchase the good, the entry is left blank. Next, your bonus is displayed. If you have not received a bonus, the value is 0. Your profit of the auction equals the sum of the two positions (value – sales price + your bonus).

Now look at the example in Figure 5. In this auction the sales price is 60, and the bonus is 10. The bidder has bought the good, and he/she gets the bonus. So his/her profit is 30.

Figure5: Screen 5



{Translation:

Sales Price:	60.0 points
Bonus:	10.0 points

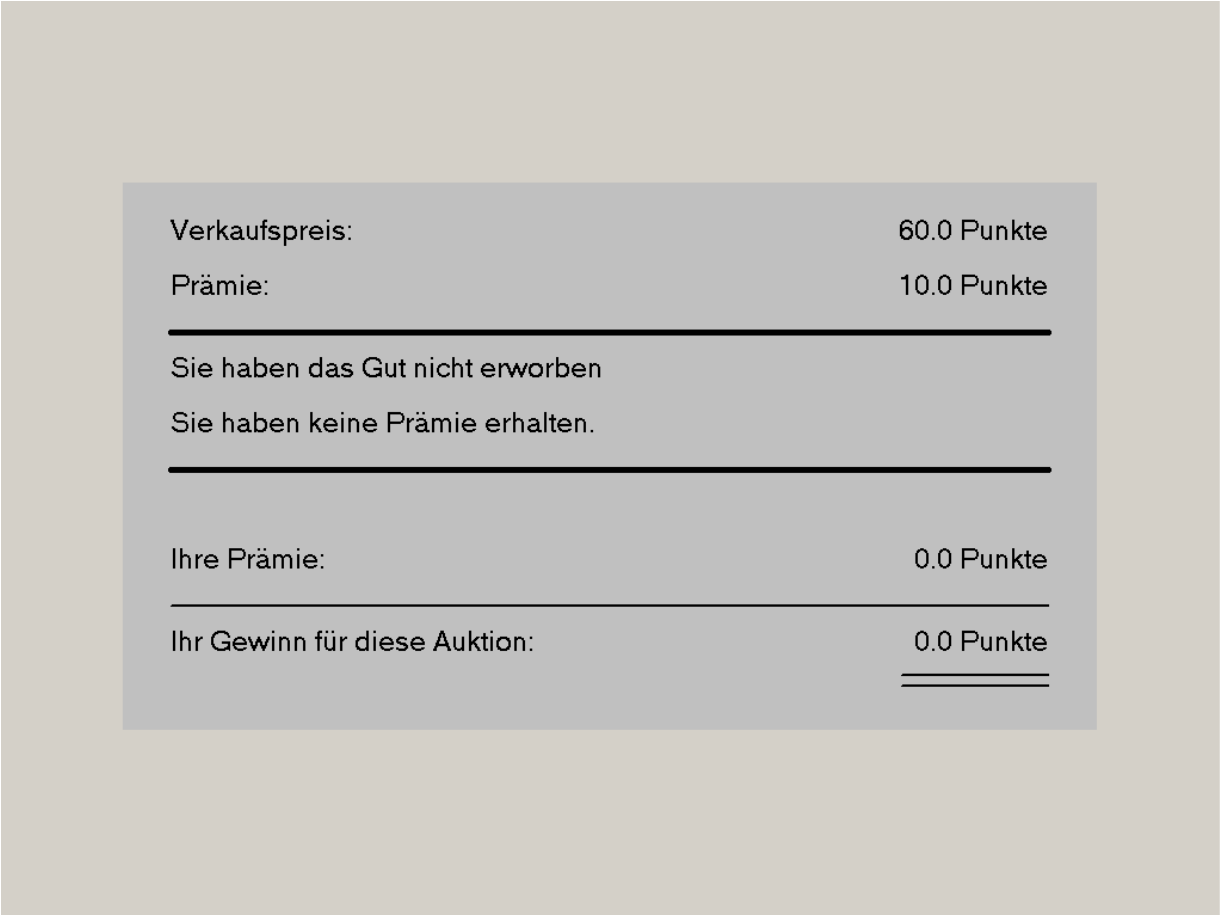
You have purchased the good at the sales price.
You have received the bonus!

Your Value – Sales Price: 80.0 – 60.0 = 20.0 points
Your Bonus 10.0 points

Your profit in this auction: 30.0 points}

In the example in Figure 6 the bidder has not purchased the good, nor received a bonus. So his/her profit is 0.

Figure 6: Screen 6



{Translation:

Sales Price: 60.0 points

Bonus: 10.0 points

You have not purchased the good.
You have not received the bonus.
Your Bonus 0.0 points

Your profit in this auction: 0.0 points}

Payment

At the end of the experiment 5 out of 15 auctions will be randomly selected. This is done by one of the participants randomly drawing without replacement 5 out of 15 balls, each marked with a number. Only the profits you gained in the 5 selected auctions will be paid. For 3 points you receive a payment of 1 euro. Additionally, you receive 5 euros. Your payment is calculated as follows:

Payment = Your profits gained in the 5 selected auctions / 3

+ 5 euros

You will receive your payment in cash at the end of the experiment. Each participant will only learn his/her own payment.

Test questions

Now you should be able to answer the test questions. Please raise your hand if you need help.

Summary

- The experiment consists of 15 auctions
- In each auction one good is sold to 1 of 4 bidders of your group
- You stay in the same group throughout the 15 auctions
- For each bidder and each of the 15 auctions, a value between 0 and 100 is selected randomly
- The price of a good is 0 at the beginning of the auction, and it is raised at regular intervals
- You can decide to drop out anytime by moving the cursor onto the blue rectangle
- When 3 or more of the 4 bidders have dropped out of your group, the auction is finished
- The bidder who did not drop out buys the good at the current price
- The basis price is the price at which the second bidder drops out
- Your profit in an auction is:
    - If you drop out first or second: 0
    - If you drop out third: bonus =(sales price – basis price)/2
    - If you buy the good: Your personal value – sales price + bonus
- At the end of the experiment, 5 auctions will be randomly selected
- Your payment is:

  Your profits gained in the 5 selected auctions / 3

  + 5 euros

Appendix C: Additional regression results

Table C1: Does the Bottom Price Affect Bids?

Dependent Variable: exit prices (bids)

	Risk Averse	Risk Seeking
Equilibrium Bid	0.36***	0.43*
	(0.064)	(0.173)
Bottom Price	0.52***	0.70***
	(0.082)	(0.092)
Constant	14.88**	-10.37
	(3.319)	(17.438)
n	74	80
R-squared	0.83	0.74
Clusters	5	5

Note: Standard errors are clustered by groups. We only use bids that were submitted after the bottom price had been established. The number of observations is not the same for the two treatments because subjects sometimes quit simultaneously.